

# Nadaka: Qualcomm NPU Verification & Benchmark Report

On-device inference on the iQOO I2501 · Snapdragon SM8850 (8 Elite Gen 5) · Hexagon HTP V81 · Android 16

iQOO Hackathon 2026, Hyderabad · measured 26 Sep 2026

## 1. Method

A dedicated benchmark screen (*BenchActivity*) runs with no camera, so nothing else competes for the NPU. For every model and backend it creates a fresh LiteRT interpreter, runs 10 warm-up inferences and then 100 timed runs on the same real 480×640 camera frame from this phone. Pre-processing, inference and post-processing are timed separately with *System.nanoTime*. The number of delegated operations is read from the runtime's own log line ("Replacing X out of Y node(s) with delegate"), not inferred from the delegate having been created. All times are milliseconds.

| Item            | Value                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------|
| Runtime         | LiteRT 1.4.2 + Qualcomm QNN LiteRT delegate 2.50.0 (HTP backend), GPU delegate, XNNPACK CPU (4 threads)   |
| Object detector | YOLOX, Qualcomm AI Hub release 0.63.0, w8a8, input uint8 1×640×640×3                                      |
| Depth model     | Depth Anything V2, Qualcomm AI Hub release 0.63.0, FP32 TFLite executed at FP16 on HTP, input 1×518×518×3 |
| Post-processing | YOLOX: dequantise + class-aware NMS. Depth: pool to 48×32 grid + floor-ruler drop-off analysis            |
| Runs            | 10 warm-up + 100 timed per configuration; statistics: median, P90, P95, min, max                          |

## 2. Verification of NPU execution

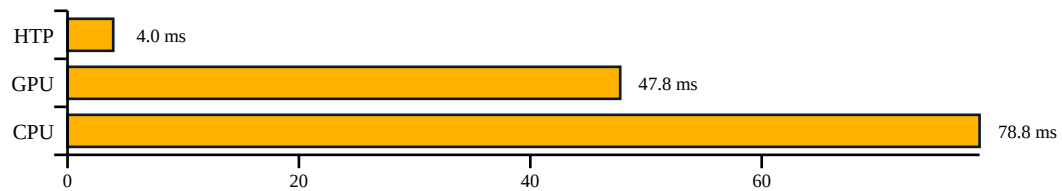
| Check                  | Result                                                                                                                                                                     |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Delegate configuration | QnnDelegate.Options: HTP_BACKEND, skel dir = nativeLibraryDir, HTP_PERFORMANCE_SUSTAINED_HIGH_PERFORMANCE, FP16 precision for depth                                        |
| HTP libraries          | libQnnHtp, libQnnHtpPrepare, libQnnHtpV75/V79/V81 Skel + Stub, libQnnSystem, libQnnTFLiteDelegate: present in the APK and in the installed native library directory        |
| YOLOX delegation       | 317/317 (TfLiteQnnDelegate)                                                                                                                                                |
| Depth delegation       | 598/598 (TfLiteQnnDelegate)                                                                                                                                                |
| Graph cache            | Was not working (no model token; only 20 B - 2 KB metadata files). Fixed with setModelToken: compiled graphs of 9.6 MB (YOLOX) and 52.7 MB (depth) now written to cacheDir |
| Debug UI               | Header shows YOLO backend + latency and Depth backend + latency separately                                                                                                 |

## 3. Summary (final build)

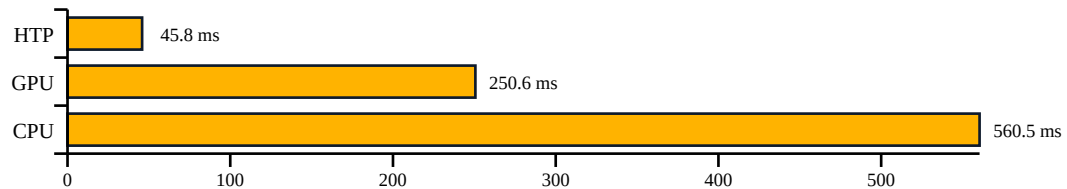
| Model             | Backend | Precision | Delegated ops | Inference median / P95    | Total median / P95 |
|-------------------|---------|-----------|---------------|---------------------------|--------------------|
| YOLOX             | CPU     | w8a8      | n/a (CPU)     | 76.34 / 88.18             | 78.83 / 90.84      |
| YOLOX             | GPU     | w8a8      | 315/317       | 35.24 / 39.82             | 47.77 / 55.07      |
| YOLOX             | HTP     | w8a8      | 317/317       | 1.78 / 1.80               | 3.95 / 4.04        |
| Depth Anything V2 | CPU     | FP32      | n/a (CPU)     | 556.47 / 587.85           | 560.47 / 592.58    |
| Depth Anything V2 | GPU     | FP32      | 598/598       | 229.66 / 232.09           | 250.64 / 256.04    |
| Depth Anything V2 | HTP     | FP16      | 598/598       | 27.50 / 28.66             | 45.84 / 50.03      |
| Depth Anything V2 | HTP     | INT8      | -             | not available (section 7) | -                  |

**Measured inference speed-up of HTP:** YOLOX 43× vs CPU and 20× vs GPU; Depth Anything V2 20× vs CPU and 8.4× vs GPU. The GPU delegate left 2 of 317 YOLOX operations on the CPU.

#### YOLOX total latency per frame (median, ms)



#### Depth Anything V2 total latency per frame (median, ms)



## 4. Optimisations applied and their measured effect

| Change                                                                          | Metric                          | Before         | After           |
|---------------------------------------------------------------------------------|---------------------------------|----------------|-----------------|
| Pre-processing: fill an array, one bulk copy (was one ByteBuffer.put per value) | YOLOX HTP pre-processing median | 15.04 ms       | 2.08 ms         |
|                                                                                 | YOLOX HTP total median          | 17.11 ms       | 3.95 ms         |
|                                                                                 | Depth HTP pre-processing median | 34.09 ms       | 10.96 ms        |
|                                                                                 | Depth HTP total median          | 63.87 ms       | 45.84 ms        |
| QNN graph cache (model token)                                                   | YOLOX HTP interpreter init      | 624 ms (cold)  | 143 ms (cached) |
|                                                                                 | Depth HTP interpreter init      | 5235 ms (cold) | 170 ms (cached) |

The delegate configuration, models, camera pipeline and tensor values are unchanged; only the host-side tensor copy and the cache token changed.

## 5. Full results: final build

| Model | Backend | Delegated ops                 | Init | Stage     | Median | P90   | P95   | Min   | Max   |
|-------|---------|-------------------------------|------|-----------|--------|-------|-------|-------|-------|
| YOLOX | HTP     | 317/317 (TfLiteQnnDelegate)   | 143  | pre       | 2.08   | 2.10  | 2.13  | 1.94  | 2.21  |
| YOLOX | HTP     | 317/317 (TfLiteQnnDelegate)   | 143  | inference | 1.78   | 1.80  | 1.80  | 1.75  | 2.07  |
| YOLOX | HTP     | 317/317 (TfLiteQnnDelegate)   | 143  | post      | 0.10   | 0.10  | 0.10  | 0.10  | 0.14  |
| YOLOX | HTP     | 317/317 (TfLiteQnnDelegate)   | 143  | total     | 3.95   | 3.99  | 4.04  | 3.82  | 4.17  |
| YOLOX | GPU     | 315/317 (TfLiteGpuDelegateV2) | 1222 | pre       | 11.00  | 14.14 | 14.48 | 3.99  | 15.21 |
| YOLOX | GPU     | 315/317 (TfLiteGpuDelegateV2) | 1222 | inference | 35.24  | 39.65 | 39.82 | 31.12 | 40.73 |
| YOLOX | GPU     | 315/317 (TfLiteGpuDelegateV2) | 1222 | post      | 1.00   | 1.20  | 1.25  | 0.26  | 1.33  |
| YOLOX | GPU     | 315/317 (TfLiteGpuDelegateV2) | 1222 | total     | 47.77  | 54.72 | 55.07 | 35.70 | 56.06 |
| YOLOX | CPU     | XNNPACK (CPU)                 | 28   | pre       | 2.27   | 2.58  | 2.64  | 2.03  | 2.93  |
| YOLOX | CPU     | XNNPACK (CPU)                 | 28   | inference | 76.34  | 87.87 | 88.18 | 75.88 | 91.42 |

| Model | Backend | Delegated ops                  | Init | Stage     | Median | P90    | P95    | Min    | Max    |
|-------|---------|--------------------------------|------|-----------|--------|--------|--------|--------|--------|
| YOLOX | CPU     | XNNPACK (CPU)                  | 28   | post      | 0.11   | 0.13   | 0.13   | 0.11   | 0.17   |
| YOLOX | CPU     | XNNPACK (CPU)                  | 28   | total     | 78.83  | 90.57  | 90.84  | 78.22  | 94.05  |
| Depth | HTP     | 598/598 (TfLiteQnnDelegate)    | 170  | pre       | 10.96  | 13.88  | 13.91  | 2.13   | 13.97  |
| Depth | HTP     | 598/598 (TfLiteQnnDelegate)    | 170  | inference | 27.50  | 28.15  | 28.66  | 26.64  | 30.85  |
| Depth | HTP     | 598/598 (TfLiteQnnDelegate)    | 170  | post      | 6.46   | 10.10  | 11.58  | 1.02   | 17.57  |
| Depth | HTP     | 598/598 (TfLiteQnnDelegate)    | 170  | total     | 45.84  | 48.73  | 50.03  | 31.78  | 50.60  |
| Depth | GPU     | 598/598 (TfLiteGpuDelegateV2 ) | 1476 | pre       | 13.98  | 14.76  | 14.92  | 4.82   | 16.52  |
| Depth | GPU     | 598/598 (TfLiteGpuDelegateV2 ) | 1476 | inference | 229.66 | 230.71 | 232.09 | 227.33 | 232.94 |
| Depth | GPU     | 598/598 (TfLiteGpuDelegateV2 ) | 1476 | post      | 8.73   | 10.18  | 10.84  | 2.45   | 12.14  |
| Depth | GPU     | 598/598 (TfLiteGpuDelegateV2 ) | 1476 | total     | 250.64 | 255.18 | 256.04 | 239.19 | 258.89 |
| Depth | CPU     | XNNPACK (CPU)                  | 89   | pre       | 2.09   | 2.26   | 2.30   | 1.91   | 3.92   |
| Depth | CPU     | XNNPACK (CPU)                  | 89   | inference | 556.47 | 581.72 | 587.85 | 548.46 | 607.28 |
| Depth | CPU     | XNNPACK (CPU)                  | 89   | post      | 1.86   | 2.01   | 2.13   | 0.65   | 2.50   |
| Depth | CPU     | XNNPACK (CPU)                  | 89   | total     | 560.47 | 585.54 | 592.58 | 551.24 | 612.49 |

## 6. Full results: baseline (before the fixes, cold cache)

| Model | Backend | Delegated ops                  | Init | Stage     | Median | P90   | P95   | Min   | Max   |
|-------|---------|--------------------------------|------|-----------|--------|-------|-------|-------|-------|
| YOLOX | HTP     | 317/317 (TfLiteQnnDelegate)    | 624  | pre       | 15.04  | 22.35 | 22.60 | 14.76 | 25.04 |
| YOLOX | HTP     | 317/317 (TfLiteQnnDelegate)    | 624  | inference | 1.95   | 2.00  | 2.03  | 1.91  | 2.25  |
| YOLOX | HTP     | 317/317 (TfLiteQnnDelegate)    | 624  | post      | 0.10   | 0.14  | 0.14  | 0.10  | 0.26  |
| YOLOX | HTP     | 317/317 (TfLiteQnnDelegate)    | 624  | total     | 17.11  | 24.39 | 24.76 | 16.80 | 27.18 |
| YOLOX | GPU     | 315/317 (TfLiteGpuDelegateV2 ) | 1626 | pre       | 20.11  | 20.20 | 20.24 | 19.91 | 24.30 |
| YOLOX | GPU     | 315/317 (TfLiteGpuDelegateV2 ) | 1626 | inference | 37.02  | 37.14 | 37.17 | 36.61 | 37.24 |
| YOLOX | GPU     | 315/317 (TfLiteGpuDelegateV2 ) | 1626 | post      | 0.14   | 0.15  | 0.16  | 0.14  | 0.18  |
| YOLOX | GPU     | 315/317 (TfLiteGpuDelegateV2 ) | 1626 | total     | 57.27  | 57.42 | 57.47 | 56.90 | 61.59 |

| Model | Backend | Delegated ops                 | Init | Stage     | Median | P90    | P95    | Min    | Max    |
|-------|---------|-------------------------------|------|-----------|--------|--------|--------|--------|--------|
| YOLOX | CPU     | XNNPACK (CPU)                 | 18   | pre       | 20.09  | 22.48  | 22.65  | 19.87  | 24.81  |
| YOLOX | CPU     | XNNPACK (CPU)                 | 18   | inference | 93.89  | 97.20  | 98.41  | 93.63  | 100.25 |
| YOLOX | CPU     | XNNPACK (CPU)                 | 18   | post      | 0.14   | 0.15   | 0.15   | 0.14   | 0.35   |
| YOLOX | CPU     | XNNPACK (CPU)                 | 18   | total     | 114.48 | 118.65 | 120.17 | 113.76 | 122.72 |
| Depth | HTP     | 598/598 (TfLiteQnnDelegate)   | 5235 | pre       | 34.09  | 34.26  | 34.32  | 33.80  | 36.39  |
| Depth | HTP     | 598/598 (TfLiteQnnDelegate)   | 5235 | inference | 27.75  | 27.87  | 28.02  | 25.95  | 28.28  |
| Depth | HTP     | 598/598 (TfLiteQnnDelegate)   | 5235 | post      | 2.07   | 3.01   | 3.31   | 0.96   | 4.67   |
| Depth | HTP     | 598/598 (TfLiteQnnDelegate)   | 5235 | total     | 63.87  | 64.95  | 65.67  | 60.98  | 66.50  |
| Depth | GPU     | 598/598 (TfLiteGpuDelegateV2) | 1574 | pre       | 81.84  | 91.65  | 95.09  | 57.09  | 102.21 |
| Depth | GPU     | 598/598 (TfLiteGpuDelegateV2) | 1574 | inference | 249.58 | 260.09 | 262.16 | 225.36 | 263.51 |
| Depth | GPU     | 598/598 (TfLiteGpuDelegateV2) | 1574 | post      | 8.72   | 9.92   | 10.76  | 0.99   | 13.12  |
| Depth | GPU     | 598/598 (TfLiteGpuDelegateV2) | 1574 | total     | 339.61 | 355.78 | 359.17 | 289.00 | 366.38 |
| Depth | CPU     | XNNPACK (CPU)                 | 97   | pre       | 34.13  | 35.80  | 36.41  | 33.84  | 42.41  |
| Depth | CPU     | XNNPACK (CPU)                 | 97   | inference | 585.02 | 594.43 | 596.69 | 577.06 | 603.32 |
| Depth | CPU     | XNNPACK (CPU)                 | 97   | post      | 1.95   | 2.11   | 2.15   | 0.68   | 2.30   |
| Depth | CPU     | XNNPACK (CPU)                 | 97   | total     | 621.23 | 630.58 | 632.84 | 611.74 | 639.82 |

## 7. Depth INT8 investigation

| Qualcomm AI Hub model (release 0.63.0)   | Float                 | Quantized                                |
|------------------------------------------|-----------------------|------------------------------------------|
| Depth Anything V2 (production)           | ONNX, QNN-DLC, TfLite | w8a16 as ONNX / QNN-DLC only (no TfLite) |
| Depth Anything (V1)                      | ONNX, QNN-DLC, TfLite | w8a16 as ONNX / QNN-DLC only (no TfLite) |
| Depth Anything V3                        | ONNX, QNN-DLC, TfLite | none                                     |
| MiDaS V2 (different model, out of scope) | ONNX, QNN-DLC, TfLite | w8a8 incl. TfLite                        |

No INT8 LiteRT build of Depth Anything V2 is published, so depth\_int8.tflite was not created and the scene-by-scene FP16 vs INT8 comparison and drop-off impact test have nothing to compare yet. Producing one would require a calibration set of recorded frames from the target scenes (flat floor, steps, stairs, kerb, ramp, shadow, puddle, shiny floor, doormat, threshold, wall, low light), plus an AI Hub account or a local ONNX to TfLite post-training quantization toolchain. Qualcomm itself ships this ViT (DINOv2) backbone at w8a16 rather than full INT8.

## 8. Recommendation

**Keep the production depth model at FP16 on HTP.** HTP inference is 27.5 ms of a 45.8 ms depth step; even a perfect 2× INT8 speed-up would save about 14 ms per depth frame, while the drop-off detector depends on sharp depth discontinuities that quantization tends to blur. Revisit only if an INT8 LiteRT model becomes available and passes the scene-by-scene drop-off test.

Limitations: one device, one input frame, room temperature, device not on a controlled thermal rig; memory usage was not measured in this run. Numbers can vary with heat and background load.